Milestone 4 - System Prototype

Project: NLP-assisted job opportunity matching for MSBA international students

Team GitHub notebook URL: https://github.com/Kongbai815/job-matching-nlp/blob/main/MSBA_Job_Matching_Milestone4_System_Prototype.ipynb

1. End-to-End System

The completed prototype accepts an advisor query, retrieves comparable real postings from an 80,000-row training corpus, predicts a four-class role-fit label, and produces a grounded recommendation. Retrieved title, company, location, skills, and label-reason fields remain visible so the advisor can verify the recommendation.

Stage	Implementation	Output
Input	Advisor query or raw posting text	Normalized text
Retrieve	Sparse inverted index over 80,000 real postings	Top-6 comparable records
Decide	Transparent MSBA role-fit rubric	high / medium / low / unclear
Generate	Evidence-bound template	Recommendation, citations, caveat
Evaluate	Balanced 4,000-row validation subset	Accuracy, macro F1, hit@5

2. Real Workflow Cases

Case	Predicted	Top retrieved evidence	Advisor use
entry level analytics	high_fit	Data Analyst (high_fit); one additional example	Forward candidates after advisor review
senior engineering filter	low_fit	Adidas Recruitment 2023 - 2+Years Experience Required - Analyst Post (medium_fit); one additional example	Hold a role that is too senior or engineering-heavy
authorization missing	unclear	Jr. Data Analyst (No OPT/CPT) (high_fit); one additional example	Escalate missing CPT/OPT or sponsorship evidence
thin metadata	unclear	ESM Data Analytics Internship - Summer 2023 (high_fit); one additional example	Investigate an incomplete posting

3. Evaluation Against Baseline

System	Evaluation set	Accuracy	Macro F1
M2 TF-IDF baseline	20,000 validation	79.3%	78.5%
Hashed centroid baseline	same 4,000 rows	79.7%	79.1%
M4 retrieval + grounded prototype	same 4,000 rows	80.0%	80.1%

On the same balanced 4,000-row subset, macro F1 rises from 79.1% to 80.1%, an absolute change of +0.009. The comparison to the 20,000-row M2 result is included for continuity but uses a different evaluation set.

4. Per-label Results and Error Analysis

Label	Precision	Recall	F1	Errors	Largest confusion
high_fit	0.930	0.993	0.960	7	unclear (6)
medium_fit	0.892	0.612	0.726	388	low_fit (380)
low_fit	0.596	0.937	0.729	63	high_fit (49)
unclear	0.979	0.660	0.789	340	low_fit (254)

The prototype is strongest on high-fit postings. Most errors occur at the medium/low and unclear/low boundaries. This is operationally important: conservative errors can hide useful opportunities, so borderline cases should be reviewed rather than automatically rejected.

5. Grounded Evidence Example

Input: Find entry-level US data analyst, business analyst, or BI roles for an MSBA student with SQL, Python, Excel, Tableau, and Power BI. Remote or Bay Area preferred.

Rank	Posting	Company / location	Label	Score
1	Data Analyst	HireMatch / Anywhere	high_fit	8.054
2	Data Analyst II, Bay Area - Remote	Astreya / Anywhere	high_fit	7.951
3	Data Analyst (US REMOTE)	LeanTaaS / Anywhere	high_fit	6.858
4	Data Analyst	HMP Global / Anywhere	high_fit	6.755
5	Data Analyst - Bay Area, Chicago, Austin, Resear	Cisco Meraki / Chicago, IL	high_fit	6.565

6. Hallucination and Governance Checks

Four workflow cases produced evidence in all four runs. Across 20 retrieved records, 1 contained an explicit authorization phrase and the saved outputs contain 0 unsupported authorization claims. The generator reports explicit text but never treats missing CPT, OPT, H-1B, or sponsorship language as evidence of eligibility.

Risk 1 is authorization hallucination; mitigation is an evidence gate and mandatory advisor escalation. Risk 2 is weak-label overconfidence; mitigation is visible evidence, no automatic rejection, and a planned human-labeled evaluation. These controls make the prototype a decision-support system rather than an automated gatekeeper.

7. Reading Connection

HOLLM Chapter 8 frames retrieval as a way to ground generated answers in an external corpus. Tunstall Chapter 7 and Jurafsky & Martin Chapter 11 emphasize answer support and extraction quality. The prototype follows that logic by separating retrieval evidence, the role-fit decision, and the authorization caveat so each claim remains inspectable.

Milestone 4 Technical Summary

Architecture and End-to-End Behavior

The prototype supports a graduate business career advisor who screens job opportunities for MSBA international students. It accepts a free-text query or posting, retrieves comparable real postings from an 80,000-row training index, applies a transparent four-label role-fit rubric, and returns a recommendation with evidence. This design uses retrieval-augmented generation as a grounding pattern: generation is a constrained template, and every job-specific claim comes from retrieved fields.

Layer	Method	Why it matters
Input	Advisor query or raw job text	Matches the real screening workflow
Retrieval	Sparse index, top-6 over 80,000 postings	Supplies comparable source evidence
Decision	Transparent role-fit rubric	Keeps the triage logic inspectable
Generation	Evidence-bound response template	Prevents unsupported job facts
Review	Advisor confirms before forwarding	Limits automation harm

Preliminary Evaluation vs. Baseline

System	Set	Accuracy	Macro F1
M2 TF-IDF baseline	20k held-out	79.3%	78.5%
Hashed centroid baseline	same 4k	79.7%	79.1%
M4 grounded prototype	same 4k	80.0%	80.1%

The same-set comparison shows a preliminary macro-F1 gain of +0.009. Retrieval succeeds for 100.0% of evaluation inputs, and support hit@5 is 90.0%. The full M2 result is context only because it uses a larger validation set.

What the Errors Mean

The prototype makes 798 errors on 4,000 balanced validation rows. High-fit F1 is 0.960, but medium-fit F1 is 0.726. The two dominant confusions are medium_fit -> low_fit (380 rows) and unclear -> low_fit (254 rows). A production pilot should therefore optimize high-fit precision and route medium/unclear cases to review instead of treating low-fit as an automatic rejection.

Grounded Case Evidence

Workflow case	Query label	Retrieved evidence pattern	Operational response
entry level analytics	high_fit	high_fit: 5	Forward candidates after advisor review
senior engineering filter	low_fit	low_fit: 2, medium_fit: 2, unclear: 1	Hold a role that is too senior or engineering-heavy
authorization missing	unclear	high_fit: 4, medium_fit: 1	Escalate missing CPT/OPT or sponsorship evidence
thin metadata	unclear	high_fit: 1, medium_fit: 4	Investigate an incomplete posting

Grounding and Hallucination Analysis

All 4 cases returned source evidence, covering 20 inspected records. One retrieved record contained explicit negative authorization language, which the system surfaced. The generated outputs contain zero unsupported authorization claims and attach an authorization verification caveat in 100% of cases.

Risk 1: authorization misinformation

The public dataset does not reliably contain CPT, OPT, H-1B, or sponsorship language. A wrong claim could cause a student to spend time on an ineligible role or miss a viable opportunity. Mitigation: do not infer eligibility, show the missing-evidence caveat, and require advisor review.

Risk 2: weak-label bias and automation harm

Labels are derived from transparent rules rather than final advisor judgments. They can over-penalize engineering vocabulary and incomplete postings. Mitigation: expose retrieved evidence, retain the human decision, report per-label errors, and collect de-identified advisor corrections for the next evaluation.

Recommendation

Use the prototype for first-pass prioritization, not automatic acceptance or rejection. Pilot it on de-identified Career Center postings, measure high-fit precision and advisor time saved, and add a separate authorization extractor only after reliable source text is available.